The AI Compute Build-Out, 2026–2030

A tracker, not a forecast

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Headline

Across the four lifecycle buckets (operational / under construction / contracted / announced), the tracker carries **56.8 GW facility power of Western capacity** and **8.6 GW of sovereign-sidebar capacity** through 2030. The split is **6.6 GW operational today**, **29.3 GW under construction** (named site, dated buildout, permits visible), **7.1 GW contracted** (counterparty + capacity disclosed via 10-K, lease, take-or-pay), and **13.9 GW announced** ("up to N GW" headlines without a contracted basis or named site). Cancellations are reported separately in §5 so they don't silently disappear from the trail.

There are no probability priors, tier-weighted central cases, or Monte Carlo distributions in this report. Every number resolves to a row in the companion `tracker.csv`, which combines Epoch AI's Frontier Data Centers (the primary upstream) with a small hand-curated overlay for capacity Epoch's documented methodology does not cover. Run `bash scripts/refresh_epoch.sh` to re-pull and rebuild.

The tracker is what it is. Every row carries a primary-source URL and a date. Aggregating across the buckets is the reader's choice; the tracker does not collapse to a single "2030 horizon" number because the underlying claims have very different evidence quality.

Bucket	Western items	Western GW	Sovereign items	Sovereign GW
Operational	21	6.586	0	0.000
Under construction	37	29.253	3	2.520
Contracted	11	7.100	3	0.314
Announced	13	13.887	9	5.810
Canceled	3	0.000	0	0.000
Total (ex-canceled)		**56.826**		**8.644**

Table 1: Bucket totals (GW facility power). Sovereign sidebar items (UAE, Saudi, India, UK, EU) are reported separately and are not added to the Western total.

Operator concentration (Western, all buckets ex-canceled)

1 Operational today

21 Western sites totaling **6.586 GW facility power** are energized as of the data cutoff (Epoch snapshot 2026-04-28). Most rows come from Epoch's Frontier Data Centers operational MW field. Wholesale-colocation sites (QTS, STACK, Vantage) appear with the AI tenant marked as undisclosed.

Operator (or operator class)	Items	GW facility
Oracle	7	8.533
Google	13	8.005
Amazon	7	7.056
Microsoft	8	5.991
Meta	8	5.308
Nebius	4	3.550
—	8	3.441
Crusoe	2	2.770
xAI	5	2.014
CoreWeave	1	1.500
Cipher Mining	2	1.300
Softbank	1	1.200
TeraWulf	1	1.000
Pantheon Atlas	1	1.000
Together AI	1	1.000
Coreweave	1	0.800
Nscale	4	0.750
Applied Digital	2	0.630
Total Western		**56.826**

Table 2: Western capacity by operator. "Wholesale colo" captures sites where Epoch tracks the facility but no AI tenant is publicly attributed. Multi-operator rows are bucketed by the lead operator.

Site	Operator	Anchor tenant	GW	low / central / high
Anthropic-Amazon New Carlisle	Amazon	Anthropic	1.092	1.092
QTS Richmond	—	—	0.854	0.854
Microsoft Fairwater Wisconsin	Microsoft	OpenAI, Microsoft	0.555	0.555
Meta Prometheus	Meta	Meta	0.531	0.531
Microsoft Fairwater Atlanta	Microsoft	OpenAI	0.433	0.433
xAI Colossus 1	xAI	xAI	0.425	0.425
Google New Albany	Google	Google DeepMind	0.407	0.407
Amazon Madison Mega Site	Amazon	Anthropic	0.341	0.341
xAI Colossus 2	xAI	xAI	0.302	0.302
OpenAI Stargate Abilene	Oracle	OpenAI	0.295	0.295
Microsoft Goodyear	Microsoft	OpenAI	0.263	0.263
Meta Temple	Meta	Meta	0.198	0.198
Google Council Bluffs (East)	Google	Google DeepMind	0.190	0.190
Google Omaha	Google	Google DeepMind	0.189	0.189
STACK Infrastructure NVA02	—	—	0.118	0.118
Google Cedar Rapids	Google	Google DeepMind	0.105	0.105
Polar (Norway) + atNorth ICE02 (Icela…	Crusoe	merchant	0.070	0.07 / 0.070 / 0.10
Fluidstack Lake Mariner	Fluidstack	Anthropic, G42	0.068	0.068
Google Pryor (North)	Google	Google DeepMind	0.065	0.065
Vantage TX1	—	—	0.045	0.045
Start Campus Sines Data Campus	Nscale	Microsoft	0.040	0.040

(continued — Operational today — Western)

Site	Operator	Anchor tenant	GW	low / central / high

Table 3: Operational today — Western

2 Under construction

37 Western sites totaling **29.253 GW facility power** are under construction. "Under construction" means a named site with dated buildout evidence: utility interconnect filings, building permits, satellite imagery, or operator-disclosed completion dates. Online column shows the central energization year from the source.

Site	Operator	Anchor tenant	GW	low / central / high	Online
Microsoft Fairwater Wisconsin	Microsoft	OpenAI, Microsoft	2.773	2.773	2027
Meta Hyperion	Meta	Meta	2.262	2.262	2028
OpenAI Stargate New Mexico	Oracle	OpenAI	2.210	2.210	2028
OpenAI Stargate Shackelford	Oracle	OpenAI	1.960	1.960	2028
QTS Cedar Rapids	—	—	1.587	1.587	2027
OpenAI Stargate Michigan	Oracle	OpenAI	1.383	1.383	2028
OpenAI Stargate Wisconsin	Oracle	OpenAI	1.300	1.300	2028
OpenAI Stargate Milam	Softbank	OpenAI	1.200	1.200	2028
xAI Colossus 2	xAI	xAI	1.192	1.192	2026
Goodnight	Google	Google DeepMind	1.088	1.088	2027
Google Fort Wayne	Google	Google DeepMind	1.035	1.035	2028
Amazon Ridgeland	Amazon	Anthropic	1.008	1.008	2027
Lebanon IN	Meta	Meta	1.000	1.000	2028-2029
Colchis (West TX) JV with AEP	Cipher Mining / AEP JV	undisclosed	1.000	0.50 / 1.000 / 1.00	2027-2028
Crusoe Abilene Expansion	Microsoft	Microsoft	0.941	0.941	2027
OpenAI Stargate Abilene	Oracle	OpenAI	0.885	0.885	2026
Coreweave Helios	Coreweave	Microsoft	0.800	0.800	2029
Meta Prometheus	Meta	Meta	0.579	0.579	2026
Google Cedar Rapids	Google	Google DeepMind	0.522	0.522	2028
Tulsa OK (Project Anthem)	Meta	Meta	0.500	0.50 / 0.500 / 1.00	2028
Amazon Madison Mega Site	Amazon	Anthropic	0.478	0.478	2026
Fluidstack Lake Mariner	Fluidstack	Anthropic, G42	0.441	0.441	2027
Microsoft Fairwater Atlanta	Microsoft	OpenAI	0.426	0.426	2026
Google Pryor (North)	Google	Google DeepMind	0.389	0.389	2026
STACK Infrastructure NVA02	—	—	0.369	0.369	2028
Lappeenranta Finland	Nebius	Nebius / multiple	0.310	0.310	2027
Google Omaha	Google	Google DeepMind	0.285	0.285	2027
Stream Phoenix	—	—	0.267	0.267	2030
Start Campus Sines Data Campus	Nscale	Microsoft	0.240	0.240	2027
Meta Kuna	Meta	Meta	0.198	0.198	2026
Anthropic-Amazon New Carlisle	Amazon	Anthropic	0.137	0.137	2026
Google New Albany	Google	Google DeepMind	0.136	0.136	2026
Vantage TX1	—	—	0.134	0.134	2027

(continued — Under construction — Western)

Site	Operator	Anchor tenant	GW	low / central / high	Online
Google Council Bluffs (East)	Google	Google DeepMind	0.094	0.094	2026
QTS Richmond	—	—	0.067	0.067	2026
Meta Temple	Meta	Meta	0.040	0.040	2026
xAI Colossus 1	xAI	xAI	0.017	0.017	2026

Table 4: Under construction — Western

3 Contracted (counterparty + capacity disclosed)

11 Western rows totaling **7.100 GW facility power** represent firm commercial commitments where a counterparty and a capacity have been publicly disclosed (typically via 10-K, executed lease, or take-or-pay press release). Site-level mapping is sometimes incomplete — the contract may name the customer and the GW figure but not the underlying physical campus. Where overlap with Epoch sites is plausible, the row range is widened to honestly bracket the dedupe uncertainty.

Site	Operator	Anchor tenant	GW	low / central / high	Online
NJ + KC fleet (sites undisclosed, mul…	Nebius	Microsoft / Meta	2.000	1.50 / 2.000 / 2.50	2027
Wholesale colo (sites undisclosed	CoreWeave	Microsoft / OpenAI / …	1.500	1.50 / 1.500 / 2.30	2027-2028
AWS sites (multiple, undisclosed)	Amazon	Anthropic	1.000	0.50 / 1.000 / 1.20	2026
Lake Mariner NY (separate from Fluids…	TeraWulf	Fluidstack / undisclo…	1.000	0.60 / 1.000 / 1.50	2027-2028
Delta Forge 1, Alexandria LA	Applied Digital	undisclosed investmen…	0.430	0.30 / 0.430 / 0.43	2027
Barber Lake TX	Cipher Mining	Amazon Web Services	0.300	0.300	2026-2027
Ward County / Cedarvale TX	Nscale	Microsoft / Google	0.240	0.24 / 0.240 / 0.94	2027
Kvandal/Narvik Norway	Nscale	Microsoft (with OpenA…	0.230	0.23 / 0.230 / 0.40	2027
Polaris Forge 2, Ellendale ND	Applied Digital	undisclosed investmen…	0.200	0.200	2026-2027
Horizon Phases 1-4 (Sweetwater TX adj…	IREN (Iris Energy)	Microsoft	0.200	0.200	2026
River Bend LA	Hut 8	Fluidstack (Anthropic…	0.000	0.24 / 0.000 / 0.24	facility

Table 5: Contracted — Western

4 Announced (headline only — no contract or named site)

13 Western rows totaling **13.887 GW facility power** central are headline-only commitments: "up to N GW" statements where no contract, no named physical site, or no executed lease is disclosed. Low cases for several rows include 0 because the underlying ceiling could lap entirely with already-counted capacity. Treat this

bucket as the speculative tail of the tracker — it is in the report because the sources are real, not because the buildout is.

Site	Operator	Anchor tenant	GW	low / central / high	Online
US sites (vast majority US per Anthro…	Google Cloud / Broadc…	Anthropic	3.500	1.00 / 3.500 / 4.20	2027
AWS sites (Project Rainier residual +…	Amazon	Anthropic	3.000	3.00 / 3.000 / 4.00	2027-2030
Cheyenne WY (Project Jade + Tallgrass…	Crusoe	Tallgrass JV / undisc…	2.700	1.80 / 2.700 / 2.70	2027-2028
Independence MO	Nebius	Nebius / multiple	1.000	0.50 / 1.000 / 1.00	2028
Topusko Croatia	Pantheon Atlas / Tran…	undisclosed hyperscal…	1.000	0.80 / 1.000 / 1.00	2029
Europe (FR, UK, IT, PT)	Together AI / Hyperte…	Together AI platform …	1.000	0.50 / 1.000 / 2.00	2027-2029
Azure sites (undisclosed)	Microsoft Azure	Anthropic	0.500	0.50 / 0.500 / 1.00	2027
Doña Ana County NM (Project Jupiter, …	Oracle	Oracle	0.500	0.50 / 0.500 / 2.45	2028-2030
near Lille FR	Nebius	Nebius / multiple	0.240	0.240	2028
Six-site US GPU cloud footprint (Puya…	Voltage Park / Lightn…	merchant	0.169	0.10 / 0.169 / 0.30	2026-2028
Loughton UK	Microsoft	Microsoft / OpenAI	0.100	0.05 / 0.100 / 0.20	2027
Kansas City + others (sites partly un…	Lambda	Microsoft / NVIDIA	0.100	0.02 / 0.100 / 0.32	2027
Memphis Colossus 2 residual / Southav…	xAI	xAI	0.078	0.08 / 0.078 / 0.26	2027

Table 6: Announced — Western

5 Canceled or paused

3 Western projects (and any sovereign cancellations) are tracked here for trail integrity. Capacity is set to zero so they don't enter the totals, but the row stays in the database with the cancellation source so future audits can see what was once announced and what didn't land.

Site	Operator	Anchor tenant	GW	low / central / high
West Texas (Project Horizon)	CoreWeave	Poolside	0.000	0.000
Cobalt Park / North Tyneside UK	OpenAI / Stargate UK	OpenAI	0.000	0.000
Abilene TX (Stargate Oracle/OpenAI 60…	Oracle	OpenAI	0.000	0.000

Table 7: Canceled / paused

6 Sovereign sidebar

State-backed AI infrastructure programmes operate under different financing and disclosure norms than the Western merchant build, so they are reported as a separate sidebar. The tracker carries **8.644 GW facility power** across 15 sovereign rows: UAE (Stargate UAE, Microsoft-G42 Khazna), Saudi Arabia (HUMAIN-xAI, HUMAIN-AMD, HUMAIN master plan residual), India (Reliance Jamnagar, Reliance Andhra, Google Visakhapatnam), the UK (Culham AI Growth Zone, BT-Nscale sovereign), and EU sovereign-AI initiatives (AMD-France LOI, Hannover Messe block).

Site	Operator	Anchor tenant	GW	low / central / high	Online
Saudi Arabia (10 additional sites in …	HUMAIN	Saudi sovereign	1.700	0.50 / 1.700 / 2.00	2028-2030
Visakhapatnam Andhra Pradesh + Bhogap…	Reliance Industries /…	India sovereign / Rel…	1.500	0.50 / 1.500 / 1.80	2028-2030
OpenAI Stargate UAE	G42	OpenAI	1.400	1.400	2028
Visakhapatnam (Vizag) India	Google (JV with Adani…	Google / Google DeepM…	1.000	0.50 / 1.000 / 1.50	2028-2030
Jamnagar Gujarat stretch beyond Phase…	Reliance Industries /…	Reliance-owned + Indi…	0.880	0.88 / 0.880 / 1.50	2028+
Saudi Arabia (first phase)	HUMAIN	xAI	0.500	0.30 / 0.500 / 0.70	2028
Germany + France (Mistral + Deutsche …	DT / Mistral / Siemens	EU sovereign / resear…	0.500	0.20 / 0.500 / 1.00	2027-2029
Saudi Arabia + US split	HUMAIN / AMD / Cisco	Saudi sovereign + Lum…	0.300	0.20 / 0.300 / 0.50	2028-2030
UAE (Khazna sites)	Khazna Data Centers /…	Microsoft Azure sover…	0.200	0.200	2026
Culham AIGZ scale-out beyond 100 MW	UK Government / UKAEA	TBD tenant	0.200	0.20 / 0.200 / 0.40	2030
Tydal Norway	Bitdeer	undisclosed (NVIDIA V…	0.180	0.180	2027
Jamnagar Gujarat Phase 1	Reliance Industries /…	Reliance-owned + Indi…	0.120	0.120	2026
Culham AI Growth Zone (initial 100 MW)	UK Government / UKAEA	TBD tenant	0.100	0.100	2027
France (sites TBD	AMD / French Govt / B…	French sovereign / re…	0.050	0.05 / 0.050 / 0.10	2027
Three BT UK sites (sub-50 MW IT each)	BT Group / Nscale / N…	UK sovereign / regula…	0.014	0.014	2027

Table 8: Sovereign sidebar — all buckets

7 Capital envelope

The bottoms-up unit-economics analysis (companion file `anatomy_layer_costs.yaml`) places one 2026-vintage AI facility GW at $30–$47 billion all-in (low to high range across the six physical layers: shell + civil, cooling, power infrastructure, networking fabric, accelerator + server BOM, grid interconnect). The paper-adopted central is $37 billion per facility-GW (rounded grid-tied). Applied to the 56.8 GW Western total ex-canceled gives a capital envelope of **$1.70 T to $2.67 T** (low to high), with a central of **$2.10 T**.

This is the build cost *implied* by the announced+contracted+under-construction Western horizon, not a

roll-up of disclosed company capex plans. It is also not a floor on what gets built, since Sightline Climate / Bloomberg analysis (April 2026, see §8) projects roughly half of the announced 2026 pipeline will slip or cancel.

8 Systemic risks

Three risks dominate the under-construction and announced buckets:

- **Grid interconnection.** Sightline Climate / Bloomberg analysis from April 2026 finds that of ~16 GW of US AI data center capacity slated for 2026 delivery, only ~5 GW is actually under active construction. Up to half of the announced 2026 pipeline faces material delay or outright cancellation. The bottlenecks are transformer and high-voltage switchgear lead times, Chinese tariff exposure on those components, and ISO interconnect queues that frequently exceed five years. Behind-the-meter generation (Crusoe Cheyenne with Tallgrass, Oracle Project Jupiter with Bloom Energy fuel cells, the new Pantheon Atlas Croatia campus with 8 GWh of battery storage) is the operational hedge against this.
- **Demand-side substitution.** The ~5 GW announced bucket is heavily concentrated in a small number of contracted-customer relationships (Anthropic across AWS / Google / Microsoft / Fluidstack; OpenAI across Oracle Stargate sites). If any single customer relationship reprices, capacity moves between operators rather than getting built. The April 2026 Anthropic–AWS expansion to up to 5 GW is the largest single mover in the 2026 calendar and post-dates the original Rev-4 cutoff.
- **Permitting and community pushback.** Several large announcements are tracking through challenged regulatory environments: Oracle Saline Township MI is under Michigan AG appeal, the OpenAI Stargate UK Cobalt Park site was paused April 9 2026 over UK regulatory and energy-cost issues, the Box Elder UT Stratos Project was tabled by commissioners over Great Salt Lake water concerns, and Wisconsin PSC rejected ratepayer cost-share for new DC power plants in early April 2026. None individually breaks the totals, but the cumulative effect tightens timelines.

9 Scope and exclusions

Three exclusions apply:

- **China.** Epoch tracks Alibaba Zhangbei (203 MW operational) but the broader Chinese AI infrastructure is materially under-disclosed in primary sources. Alibaba's August 2025 quarterly capex of $5.4 B (38.6 B yuan) and the February 2025 three-year 380 B-yuan (~$53 B) commitment are within an order of magnitude of the smaller US hyperscalers in dollar terms but cannot be corroborated at site granularity. China is excluded from Western totals; the qualitative gap is acknowledged.
- **Sub-100 MW sites.** Epoch's documented coverage threshold is roughly 100 MW. Smaller neocloud and edge deployments (Voltage Park's six-site fleet at ~169 MW total, Together AI's three-site operational footprint, the BT-Nscale UK sovereign 14 MW) appear in the overlay where the operator and aggregate capacity are public, but many smaller deployments below the threshold are not tracked at all.
- **Pre-permit speculation.** Multi-gigawatt visions without permits, financing, or anchor tenants are excluded. Fermi America's 11–17 GW Amarillo TX vision is the canonical case — anchor tenant pulled out in April 2026, class action filed, stock down 46% — and stays excluded.

10 Methodology

Epoch AI's Frontier Data Centers dataset (CC BY 4.0, snapshot 2026-04-28) is the primary upstream. Epoch tracks 37 frontier sites with their own `#confident` / `#likely` / `#speculative` confidence tags and publishes confidence bands of ×1.4 on power, ×1.5 on compute, ×1.6 on cost, and ±6 months on timing. Their methodology is documented at epoch.ai/data/data-centers-documentation.

This report adds a small overlay (`overlay.csv`, 46 rows) for capacity Epoch's documented methodology does not cover: ceiling-style "up to N GW" announcements without site mapping (Anthropic–AWS 5 GW, Anthropic–Google/Broadcom 3.5 GW, Anthropic–Azure 1 GW), contracted neocloud fleet ex-Epoch (CoreWeave 10-K, Nebius press, Applied Digital lease, Hut 8 / Cipher / TeraWulf / IREN backlogs), sub-threshold or non-US sites pending Epoch index expansion (Crusoe Nordic, BT-Nscale UK), and the sovereign sidebar.

The pipeline is two scripts. `scripts/refresh_epoch.sh` pulls fresh data and rebuilds; `scripts/build_tra` combines Epoch's compiled JSON with the overlay CSV into `tracker.csv` and `tracker.md`. There is no probability framework, no realization-priors model, no Monte Carlo draw, and no deterministic tier-weighted central case. The four buckets describe lifecycle stage; the reader does the aggregating.

Audit basis. Three independent deep-research passes (Anthropic, OpenAI, Google Gemini) audited the prior database against primary sources between 2026-04-27 and 2026-04-29. Their findings are baked into the current overlay row magnitudes and ranges, especially around the Anthropic–AWS expansion to 5 GW, the Broadcom 8-K disclosing 3.5 GW of TPU compute commitment, the Reliance Andhra upsizing to 1.5 GW following the April 25 Andhra Pradesh IPC approval, the Crusoe Nordic operational figure correction, and the Stargate Abilene Oracle/OpenAI 600 MW expansion replacement by Crusoe-Microsoft.

Sources and references

Every row in `tracker.csv` carries a `source_url` and a `source_date`. Companion files: `overlay.csv` (the hand-curated rows that combine with the Epoch upstream), `anatomy_layer_costs.yaml` (capex unit-economics, six layers × ~40 sub-components, vintage-tagged), `anatomy_named_projects.yaml` (disclosed project-level capex anchors: Crusoe Abilene, Meta Hyperion, AEP/SB Energy Piketon, Microsoft Mt Pleasant, AWS Madison, Talen-AWS), `forecaster_capex_comparison.yaml` (cross-forecaster reconciliation across 18 sources: Bernstein, Bain, JLL, Cushman, Goldman, Morgan Stanley, JPM, BNEF, Synergy, Dell'Oro, IEA, LBNL, Aschenbrenner, NVIDIA, Barclays, Stargate, Hyperion, CoreWeave, Nebius). The eight per-layer research notes live under `research/`.